

Spotify Audio Features Dataset

Overview

Attribute	Value
Source	Kaggle
Samples	114,000 tracks
Audio Features	9 characteristics
Genres	114 categories
API	Spotify Web API
Missing Values	Some rows may have missing values; check before analysis

Description

This dataset contains audio features for over 114,000 tracks from Spotify, spanning 114 different genres. The audio features are computed by Spotify's proprietary algorithms and describe various acoustic characteristics of each track.

Audio features are algorithmically extracted summaries of the raw audio signal. They are designed to capture distinct perceptual and acoustic properties of music, making them useful for tasks like genre classification, playlist generation, and music recommendation.

Audio Features (Analysis Variables)

Feature	Description	Range
danceability	How suitable a track is for dancing, based on a combination of tempo, rhythm stability, beat strength, and overall regularity	0.0–1.0
energy	A perceptual measure of intensity and activity; energetic tracks feel fast, loud, and noisy	0.0–1.0
loudness	Overall loudness of the track in decibels (dB), averaged across the entire track	approx. -60 to 0 dB
speechiness	Detects the presence of spoken words; values above 0.66 are likely entirely spoken word, 0.33–0.66 may contain music and speech	0.0–1.0
acousticness	A confidence measure of whether the track is acoustic (i.e., not electronically amplified)	0.0–1.0
instrumentalness	Predicts whether a track contains no vocals; values closer to 1.0 indicate greater likelihood of no vocal content	0.0–1.0
liveness	Detects the presence of an audience in the recording; values above 0.8 suggest the track was performed live	0.0–1.0
valence	A measure of the musical positiveness conveyed; high valence sounds more positive (happy, cheerful, euphoric), low valence sounds more negative (sad, depressed, angry)	0.0–1.0
tempo	The estimated overall tempo of a track in beats per minute (BPM)	variable (typically 50–250 BPM)

Note on scales: Most features are normalized to $[0, 1]$, but loudness is on a decibel scale and tempo is in BPM. Consider whether standardization is needed for your analysis.

Additional Metadata

Column	Description
track_id	Spotify track ID
artists	Artist name(s)
album_name	Album name
track_name	Track title
popularity	Popularity score (0–100, based on recent play counts)
duration_ms	Track length in milliseconds
explicit	Explicit content flag (True/False)
track_genre	Genre category (one of 114 genres)

Data Access

Loading from Local CSV (Recommended)

The dataset is provided as `dataset.csv` in this folder.

Python:

```
import pandas as pd

df = pd.read_csv('dataset.csv')

audio_features = ['danceability', 'energy', 'loudness', 'speechiness',
                  'acousticness', 'instrumentalness', 'liveness', 'valence', 'tempo']
X = df[audio_features]
genre = df['track_genre']

print(f"Tracks: {len(df)}")
print(f"Audio features: {len(audio_features)}")
print(f"Genres: {genre.nunique()}")
```

R:

```
df <- read.csv('dataset.csv')

audio_features <- c('danceability', 'energy', 'loudness', 'speechiness',
                  'acousticness', 'instrumentalness', 'liveness', 'valence', 'tempo')
X <- df[, audio_features]
genre <- df$track_genre

cat("Tracks:", nrow(df), "\n")
cat("Audio features:", length(audio_features), "\n")
cat("Genres:", length(unique(genre)), "\n")
```

Julia:

```
using CSV, DataFrames

df = CSV.read("dataset.csv", DataFrame)

audio_features = [:danceability, :energy, :loudness, :speechiness,
```

```
        :acousticness, :instrumentalness, :liveness, :valence, :tempo]
X = select(df, audio_features)
genre = df.track_genre

println("Tracks: $(nrow(df))")
println("Audio features: $(length(audio_features))")
println("Genres: $(length(unique(genre)))")
```

Data Source & Attribution

Data was collected using the Spotify Web API. Spotify owns all intellectual property rights to the audio features and metadata.

Kaggle Dataset: <https://www.kaggle.com/datasets/maharshipandya/-spotify-tracks-dataset>

Spotify API Documentation: <https://developer.spotify.com/documentation/web-api/reference/get-audio-features>

Terms of Use

This data is intended for educational and research purposes. Commercial use may require authorization from Spotify. The audio features are computed by Spotify's proprietary algorithms.

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Documentation written by Claude (Anthropic), February 2026